

Improving Uncertainty Communication in Scientific Visualization*

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Abstract

Scientific visualization frequently presents point estimates while undercommunicating the uncertainty behind them, which can lead viewers to draw overconfident conclusions. This study compares three widely used uncertainty encodings (error bars, violin + box plots, and a *static* hypothetical outcome plots (HOPs) grid) in a within-subjects experiment with 16 participants using the Palmer Penguins dataset. Parallel question sets and arithmetic distractors between blocks were used to reduce carry-over effects. In this sample, the HOPs grid had the highest aggregate accuracy (100% of 48 trials), followed by error bars (97.9%) and violin/box plots (91.7%). All three encodings were at or near ceiling, so the 100% figure should be read as “no errors observed in this small, high-data-literacy sample on these specific tasks,” not as evidence that HOPs are universally perfect. A participant-level, tie-corrected Friedman test on accuracy was significant ($\chi^2 = 6.50$, $df = 2$, $p = 0.039$, Kendall's $W = 0.20$), while response time differences were not ($\chi^2 = 4.63$, $df = 2$, $p = 0.099$). Post-hoc pairwise Wilcoxon tests did not survive Bonferroni correction, so these results are best read as preliminary signal rather than conclusive evidence. They are consistent with the possibility that showing multiple plausible outcomes helps viewers make better judgments on simple comparison tasks, and they motivate a larger follow-up study with harder stimuli.

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1 Introduction

Scientific visualizations routinely present results that come from measurement, modeling, or sampling, all of which carry uncertainty. When that uncertainty is left out or poorly conveyed, viewers can draw overconfident conclusions or misjudge how strong the evidence really is [1]. The trouble is not a lack of encoding options: error bars, box plots, violin plots, and hypothetical outcome plots (HOPs) are all readily available in modern charting libraries. The trouble is that we do not have great empirical guidance on which option actually helps people the most, especially when the goal is not just “get the right answer” but also “know how sure you should be about it.”

This study runs a controlled, within-subjects comparison of three widely used uncertainty encodings on three standardized task types. Each task has a pre-specified ground truth: a *ranking* task (pick the species with the largest/smallest/largest mean bill length), an *estimation* task (report a species’ mean bill length to within ± 2 mm), and a *decision* task (agree or disagree with a claim about how two species compare). Each participant completes one such block of three tasks per encoding. We measure decision accuracy, confidence calibration, and response time. The dataset used for all stimuli is the Palmer Penguins dataset [2], chosen because it is small, clean, well-documented, and has three species with means that are close enough to require real attention to the uncertainty (Adelie = 38.8 mm, Gentoo = 47.5 mm, Chinstrap = 48.8 mm) yet far enough apart to define an unambiguous ground truth. The Gentoo versus Chinstrap comparison in particular is the kind of case where a reader looking at a point estimate alone might over-commit, which is exactly the situation uncertainty encodings are designed to support. For HOPs, we use a *static* grid of twelve bootstrap resamples rather than the animated HOPs form typical in the original literature [3], because a static display is what most practitioners can actually embed in a paper, slide deck, or dashboard.

1.1 Motivation

Surveys like Mehta [1] map how uncertainty enters the visualization pipeline and catalog the encoding strategies people use, but they stop short of recommending one strategy over another because the empirical evidence is thin. Petek et al. [4] offer a formal framework (treat the visualization as a function of uncertain inputs and propagate that uncertainty into a distribution over images), but their focus is on the math, not on whether end users actually benefit. On the empirical side, Stokes et al. [5] compare speech, text, and visualization as modalities for communicating uncertainty and find that visualization supports rational decision-making, but they do not break the “visualization” category down further into encoding types. The gap between

knowing visualization works and knowing which encoding works best is what we are trying to fill.

Applied work in scientific visualization reinforces this from a different angle. Saklani et al. [6] show that adding uncertainty overlays to DNN-based volume renderings increases trust among domain scientists, and Li et al. [7] design bespoke uncertainty tubes for particle trajectories. These encodings are tightly coupled to the structures they visualize (volumetric scalar fields, particle paths); their design lessons do not straightforwardly transfer to the bar-chart-style group comparisons that dominate day-to-day scientific communication. When a practitioner has a handful of sample means and wants to convey uncertainty around them, they typically reach for a general-purpose encoding: error bars, a violin or box plot, or some form of HOPs. The empirical question of which of those general-purpose encodings best supports accurate, well-calibrated judgments is largely unsettled in the recent literature, which is the gap our study addresses.

1.2 Research Questions

1. **RQ1 (accuracy).** Which encoding produces the highest proportion of correct responses across the three task types, where *correct* is defined against a pre-specified ground truth for each trial (species identity for ranking, numeric answer within $\pm 2\text{mm}$ for estimation, agree/disagree match for decision)?
2. **RQ2 (calibration).** Which encoding produces the best alignment between stated confidence and actual correctness, as measured by the Brier score $(\text{confidence}/5 - \text{correct})^2$ averaged per condition, with an overconfidence index $(\text{mean confidence}/5 - \text{mean accuracy})$ reported alongside?
3. **RQ3 (time cost).** Do distributional encodings incur a measurable cost in per-trial completion time compared to error bars, where completion time is measured from stimulus render to response submission?

1.3 Hypotheses

- **H1 (maps to RQ1).** Distributional encodings (violin/box and HOPs) will yield higher accuracy than error bars. HOPs in particular should do well on ranking and decision tasks because each panel makes the between-group difference visually explicit.
- **H2 (maps to RQ2).** Violin/box will yield the best Brier-score calibration because the continuous density shape maps more naturally onto “how sure

should I be?” judgments, while HOPs may leave participants uncertain about how to aggregate over discrete draws.

- **H3 (maps to RQ3).** Both distributional encodings will take longer per trial than error bars, but by no more than roughly 15 seconds on average. This threshold was set a priori by budgeting: the full 9-trial survey was advertised to participants as a ~ 10 -minute task, which leaves a per-block budget of about 3 minutes; a per-trial overhead above ~ 15 seconds would push a distributional block past that block budget and become a noticeable cost rather than a marginal one.

The pilot run ($n = 4$) and the main study ($n = 16$) tell partially different stories, which is worth flagging up front. In the pilot, error bars and violin/box plots had similar accuracy and timing, while HOPs were slower and slightly less accurate, which we attributed to unfamiliarity with the static grid format. Based on the pilot we made three changes before the main study: we added parallel task sets (three different ranking targets, three different estimation species, and three different decision claims) to reduce answer carry-over, inserted a short arithmetic distractor between encoding blocks, and added an end-of-study recognition check. In the main study, accuracy rose across all three encodings (see Results), HOPs moved from worst to best on accuracy, and the accuracy gap between encodings narrowed enough that a ceiling effect is a real concern. H1 is partly supported by the main-study data; H2 and the exact timing claim in H3 are not.

1.4 Scope and Organization

Section 2 reviews related literature and identifies the empirical gap this study addresses. Section 3 describes the methodology, sample, stimuli, and analysis plan. Section 4 reports results for accuracy, response time, and calibration. Section 5 discusses findings, implications, limitations, and future work. Section 6 concludes.

2 Literature Review

2.1 Uncertainty in the Visualization Pipeline

Mehta [1] breaks the visualization pipeline into four stages (data collection, preprocessing, visualization, and inference) and surveys how uncertainty enters at each one. The key takeaway for our study is that even when the data itself is clean, the visualization stage introduces representational choices that affect how viewers interpret uncertainty. A bar chart with error bars communicates something different than a violin plot of the same data, and both communicate

something different than a grid of plausible outcomes. Mehta’s survey calls for more empirical work measuring how these choices affect viewer judgment.

2.2 Formal Frameworks for Uncertainty Encoding

Petek et al. [4] propose treating a visualization as a function of uncertain inputs. If the inputs carry a probability distribution, the output is a distribution over images rather than a single image. Familiar representations like confidence intervals and prediction bands fall out naturally from this framework. We use a simplified version of their resampling approach to generate the HOPs stimuli: for each “draw,” we resample the raw data with replacement and plot the resulting species means.

2.3 Modality Comparisons

Stokes et al. [5] run two crowdsourced experiments comparing speech, text, and visualization for communicating data uncertainty. Visualization and text supported rational decision-making, while speech inspired more trust but also riskier choices. Their study measures both decision quality and confidence, which we adopt. The limitation is that “visualization” is treated as a single category; they do not vary the encoding within it. We extend their approach by holding the modality constant and varying the encoding.

2.4 Uncertainty in Scientific Visualization Applications

Saklani et al. [6] tackle uncertainty-aware volume rendering using Deep Ensembles and Monte Carlo Dropout, showing that surfacing prediction uncertainty makes DNN-based visualizations more trustworthy. Li et al. [7] work on particle trajectories with an “uncertainty tube” that captures nonsymmetric error. Both demonstrate that domain-specific encodings help, but neither tests the general-purpose encodings that most researchers reach for when they want to add uncertainty to a standard plot.

2.5 Research Gap

The literature shows that uncertainty is pervasive, formal tools exist for encoding it, and the choice of representation matters. What is missing is a controlled head-to-head comparison of three widely used general-purpose encodings on the same standardized tasks, with both accuracy and calibration measured explicitly. By *accuracy* we mean the proportion of task responses that match the pre-specified ground truth for that trial; by *calibration* we mean the agreement between a participant’s stated confidence (1–5 Likert, mapped to [0.2, 1.0]) and whether their answer was actually correct, quantified by the Brier score

and supplemented with an overconfidence index (mean confidence/5 − mean accuracy). The present study addresses that gap directly.

3 Methodology

3.1 Research Design

We used a within-subjects experimental design: every participant saw all three encodings. Encoding order was counterbalanced by randomly shuffling the three conditions per session, giving each participant one of six possible orderings drawn uniformly at random. Task order within each block was fixed (ranking, estimation, decision).

A brief arithmetic distractor task appeared between each encoding block to break short-term retention. At the end of the session, participants answered a recognition-check question to provide self-report data on whether carry-over was still present. This yields 9 trials per participant (3×3), for a total of 144 trial observations.

3.2 Participants

Sixteen participants were recruited through convenience sampling, mostly friends and classmates from a range of departments. All had normal or corrected-to-normal vision and were comfortable reading English. The sample skewed young (8 aged 18–24, 6 aged 25–34, one 35–44, one 45+) and technically literate (self-rated data literacy: 3 at level 3, 4 at level 4, 9 at level 5; median = 5). Fields of study included Computer Science (4), Mechanical Engineering, Electrical Engineering, Data Science, Bioengineering, Physics, Mathematics, Psychology, Symbolic Systems, Business Administration, Interaction Design, Computational Health, and Chicanx/Latinx Studies (1 each). Participation was voluntary and anonymous; no compensation was offered.

3.3 Materials

Stimuli

Three chart types were generated from the Palmer Penguins dataset [2], each showing bill length (mm) by species (Adelie, Chinstrap, Gentoo):

1. **Error bars:** bar chart of species means with 95% confidence interval error bars. This is the baseline condition.
2. **Violin + box plot:** violin plot overlaid with a box plot showing distribution shape, median, and quartiles.

3. **HOPs grid:** a 3×4 grid of 12 bootstrap draws, each panel showing re-sampled species means as dot/lollipop plots, with a caption explaining the display.

All stimuli were generated using matplotlib 3.9.4 and are available in the project repository.

Tasks

To prevent answer carry-over between encoding blocks, we used three parallel task sets with structurally identical question types but different content:

- **Set 1 (Block 1):** Ranking: which species has the *largest* mean (Chinstrap). Estimation: estimate the mean for Chinstrap (48.8 mm). Decision: claim that Gentoo and Chinstrap have the same mean (Disagree).
- **Set 2 (Block 2):** Ranking: which species has the *smallest* mean (Adelie). Estimation: estimate the mean for Gentoo (47.5 mm). Decision: claim that Adelie and Gentoo have the same mean (Disagree).
- **Set 3 (Block 3):** Ranking: which species has the *second-largest* mean (Gentoo). Estimation: estimate the mean for Adelie (38.8 mm). Decision: claim that Chinstrap has a larger mean than Gentoo (Agree).

Correctness was defined per task type:

- *Ranking:* the selected species matches the known largest, smallest, or second-largest mean for that block.
- *Estimation:* the numeric answer is within ± 2 mm of the true species mean. The ± 2 mm tolerance was chosen a priori to be close to one standard error of the species-level means at the dataset’s sample sizes ($n \approx 50$ –150 per species), giving enough slack to absorb minor arithmetic or reading error but still tight enough to discriminate among Adelie (38.8 mm), Gentoo (47.5 mm), and Chinstrap (48.8 mm).
- *Decision:* the agree/disagree response matches the ground-truth relation between the two referenced species.

Confidence was rated on a 1–5 Likert scale immediately after each task.

Platform

Self-contained HTML/CSS/JS survey page with static PNG stimuli. Response timing was captured via JavaScript `performance.now()`. Results were displayed as JSON for manual collection via copy-to-clipboard and download buttons.

3.4 Procedure

1. Consent page with study description and time estimate (~ 10 min).
2. Demographics: age range, major/field, self-rated data literacy (1–5).
3. Instructions.
4. Nine trials in three blocks. A short arithmetic distractor appeared between blocks.
5. Recognition check.
6. Open-ended feedback.
7. Thank-you page with JSON export (copy or download).

Most participants were walked through the survey over FaceTime or Zoom; a few used the hosted link independently.

3.5 Measures

- **Decision accuracy:** proportion of correct responses per encoding per participant, then averaged across participants for the condition-level summary.
- **Confidence calibration:** Brier score, $(\text{confidence}/5 - \text{correct})^2$, averaged per condition. Lower is better. Confidence is measured on a 1–5 Likert scale, which maps under this scheme to probabilities in $[0.2, 1.0]$, not $[0, 1]$. That has two implications worth calling out: (a) the Brier score cannot reach 0 for a participant who answers incorrectly even at the minimum confidence value, and (b) the 5-step Likert scale is coarse, so a participant who chose “3” could plausibly have meant anywhere from about 55% to 65% confidence given finer granularity. We therefore pair Brier with an **overconfidence index**, defined per condition as $(\text{mean confidence}/5) - (\text{mean accuracy})$. Positive values indicate overconfidence, negative values indicate underconfidence.
- **Response time:** milliseconds from the moment the stimulus and question text finished rendering on the participant’s screen until they clicked Submit for that trial, captured via JavaScript `performance.now()`. This interval does not include time spent on the consent page, demographics, instructions, or the between-block arithmetic distractors, and it does not include time spent locating or loading the survey itself. It is intended to reflect just the interpretation cost of the encoding on that trial.

3.6 Analysis

All statistical tests were conducted at the **participant level** ($n = 16$), not the trial level ($n = 144$). For each participant and each encoding, we first computed the mean accuracy and the mean response time across that participant’s three trials in the block, yielding three paired values per participant per metric.

For the overall across-encoding comparison we used the Friedman test with tie correction, which is the appropriate non-parametric repeated -measures test for three related samples. We report Kendall’s coefficient of concordance W as a standardized effect size ($W \in [0, 1]$). Where the omnibus Friedman test was significant we ran post-hoc pairwise Wilcoxon signed-rank tests (two-sided, normal approximation with continuity correction) on the three encoding pairs and applied Bonferroni correction ($\alpha = 0.05/3$). We also report per-task-type breakdowns and an IQR-based outlier check on the single sub-cell that fell below ceiling (violin/box ranking).

An earlier version of this analysis computed Friedman ranks without handling ties in participant-level accuracy scores (many participants had tied condition-level accuracies at 100%), which substantially inflated the chi-square statistic. The numbers reported in this draft use the tie-corrected implementation.

3.7 Ethical Considerations

This study was not submitted to an IRB for review. Its design is consistent with the CSU Chico exempt-category criteria for anonymous, minimal-risk educational research (no PII, voluntary participation, ability to withdraw at any time), but the exempt classification was self-assigned by the researcher rather than formally granted. Participants provided informed consent on the first page, no data were collected from minors, and all session exports were stored locally and not transmitted to any external server.

4 Results

4.1 Decision Accuracy

In this sample of 16 participants and 48 trials per encoding, the HOPs grid had the highest aggregate accuracy ($48/48 = 100\%$), followed by error bars ($47/48 = 97.9\%$) and violin/box ($44/48 = 91.7\%$). We use “100%” here in its narrow, descriptive sense: zero errors observed on these three task types, by these sixteen participants, on this one clean dataset. It is not a claim that HOPs produce error-free judgments in general. Nine of the sixteen participants were themselves at 100% on all three encodings, and every participant was at

ceiling on the estimation and decision task types for error bars and HOPs. This compresses what the statistical test can see: a tie-corrected Friedman test at the participant level was significant but with a modest effect size ($\chi^2 = 6.50$, $df = 2$, $p = 0.039$, Kendall’s $W = 0.203$), and none of the three post-hoc Wilcoxon signed-rank tests cleared the Bonferroni-corrected threshold of $\alpha = 0.017$ (error bars vs. violin/box $p = 0.18$, error bars vs. HOPs $p = 1.00$, violin/box vs. HOPs $p = 0.10$). Almost all of the between-encoding signal came from a single cell: the violin/box *ranking* task, where accuracy was 81.2%. An IQR-based check on that cell flagged three participants as outliers; those three participants accounted for *every* ranking error in the violin/box condition. The other thirteen participants were at ceiling on that task. Estimation accuracy was 100% in all three encodings, which makes that task uninformative for separating the conditions. Given the high ceiling, the small sample, and the fact that the omnibus test does not survive a conservative correction, the accuracy result should be read as preliminary rather than conclusive.

Table 1: Decision accuracy by encoding and task type ($n = 16$ participants, 48 trials per encoding). Cells at 100% reflect zero errors observed in this sample on these specific tasks, not a universal claim about the encoding.

Encoding	Overall	Ranking	Estimation	Decision
Error bars	97.9%	100%	100%	93.8%
Violin/box	91.7%	81.2%	100%	93.8%
HOPs grid	100%	100%	100%	100%

Figure 1 shows the accuracy breakdown with 95% confidence intervals. The violin/box ranking condition is the only cell below ceiling.

4.2 Response Time

Violin/box had the longest average response time (39.9 s). Error bars (24.3 s) and HOPs (26.1 s) were comparable. The Friedman test on response time was not significant at $\alpha = 0.05$: $\chi^2 = 4.63$, $df = 2$, $p = 0.099$, Kendall’s $W = 0.145$ (a small effect). Descriptively, violin/box was about 15.6 s slower per trial than error bars and 13.8 s slower than HOPs, but the within-participant pattern was inconsistent enough that the omnibus test did not cross significance, and we did not run post-hoc comparisons for that reason. A larger sample might resolve this into a significant effect, but we do not claim one here.

Figure 2 shows the distribution of response times by encoding. Violin/box had the widest spread and highest median, though the Friedman test was not significant.

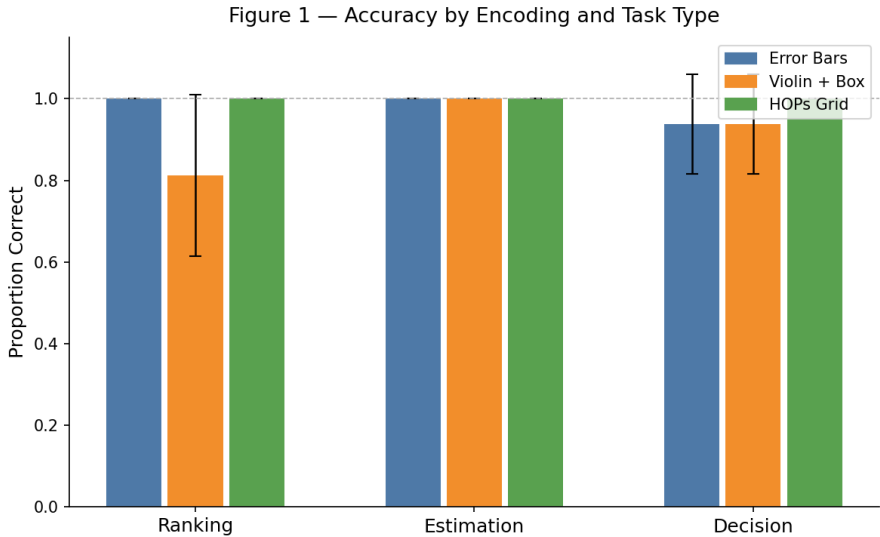


Figure 1: Proportion correct by encoding and task type. Error bars show 95% confidence intervals. Violin/box ranking is the only condition below ceiling, and the bars at 1.0 should be read as “no errors observed in this small sample,” not as a universal accuracy claim.

4.3 Confidence and Calibration

Mean confidence ratings were similar across encodings: error bars (4.65/5), HOPs grid (4.50/5), and violin/box (4.42/5). Brier scores were low across the board, with error bars showing the best calibration (0.025) and violin/box the worst (0.048). The overconfidence index was slightly *negative* in every condition (error bars -0.050 , violin/box -0.033 , HOPs -0.100), meaning participants were, if anything, slightly underconfident relative to their near-perfect accuracy. This is the expected pattern when accuracy is pressed against a 1.0 ceiling: stated confidence of $4/5$ ($= 0.8$) still leaves a negative index when accuracy is 1.0. The narrow Brier spread (0.025–0.048) should therefore be interpreted against this ceiling, not as strong evidence that one encoding is meaningfully better calibrated than another.

Figure 3 provides a per-participant view of trial-level correctness alongside each participant’s age range and self-rated data literacy.

Table 2: Mean response time (seconds) by encoding and task type.

Encoding	Overall	Ranking	Estimation	Decision
Error bars	24.3	23.8	25.9	23.2
Violin/box	39.9	40.6	41.7	37.4
HOPs grid	26.1	21.6	37.4	19.4

Table 3: Confidence and calibration by encoding. The overconfidence index (OC) is mean confidence/5 minus mean accuracy; negative values indicate underconfidence.

Encoding	Mean conf. (1–5)	Brier	OC index
Error bars	4.65	0.025	−0.050
Violin/box	4.42	0.048	−0.033
HOPs grid	4.50	0.030	−0.100

5 Discussion

5.1 Interpretation of Results

Our main-study accuracy pattern did not line up with H1 and H2 in the way we expected: HOPs, not violin/box, led on accuracy, and violin/box was actively worse than error bars on the ranking task. A few non-exclusive explanations are consistent with the data:

- *Task-encoding alignment.* Each HOPs panel makes the between-group ordering visible as a small picture; a participant doing a ranking task only has to scan panels and count “which species is higher, most of the time.” That is close to an optimal match between task and display.
- *Density dominance in violin/box.* The violin’s bulbous density silhouette is visually louder than the thin box-plot median line it wraps around. On ranking, where the relevant cue is a single central tendency, the density shape seems to have pulled attention away from the median, leading the three outlier participants to rank by visual mass rather than by the embedded summary statistic.
- *Unfamiliarity as an attention cue.* Participants had far more prior exposure to error bars than to HOPs, so the HOPs condition may have forced slower, more deliberate looking, while the familiar error bars may have encouraged a quicker pattern match. Unfamiliarity can paradoxically improve accuracy on simple tasks by suppressing overlearned heuristics.
- *Cognitive load and ceiling effects.* All three task types had a clear ground

Figure 3 — Response Time Distribution by Encoding

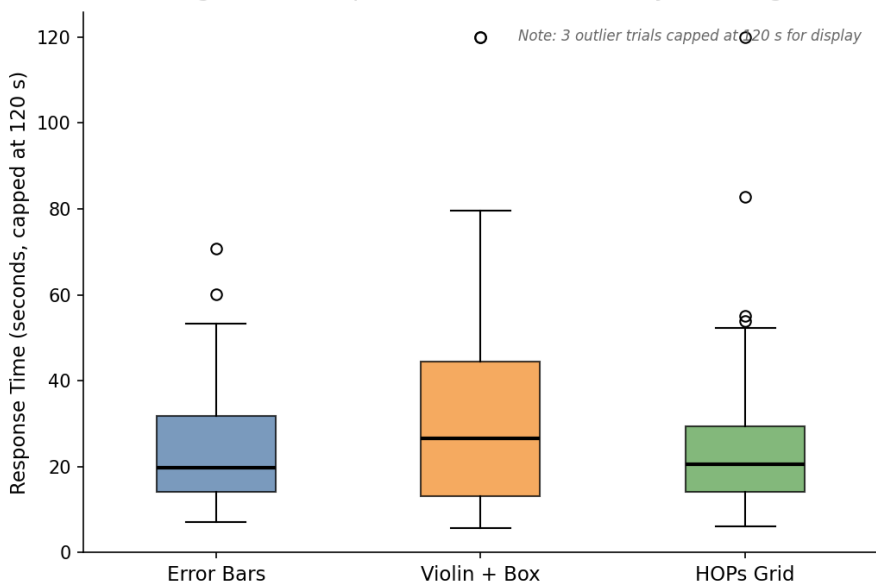


Figure 2: Response time distributions by encoding. Boxes show the interquartile range; whiskers extend to $1.5 \times \text{IQR}$.

truth and a clean, well-separated dataset; the tasks were arguably too easy to discriminate the encodings, which is consistent with the high overall accuracy and the compressed Brier-score range.

Error bars had high accuracy (97.9%) and the fastest response times (24.3 s), with the best numerical Brier score (0.025). We are cautious about calling them the best-calibrated encoding because, as noted, the ceiling in accuracy mechanically compresses Brier score across conditions; the ordering among 0.025, 0.030, and 0.048 is not something we would bet on outside this specific task set.

5.2 Calibration

H2 (violin/box would produce the best calibration) was not supported. Error bars had the numerically lowest Brier score, violin/box the highest, and all three encodings sat in a narrow band where the overconfidence index was slightly negative. The most honest reading is that accuracy near 1.0 mechanically caps how much Brier score can vary between conditions. A harder task

Figure 10 — Trial-by-Trial Correctness Heatmap
(green = correct, red = wrong; rows sorted by collection time)

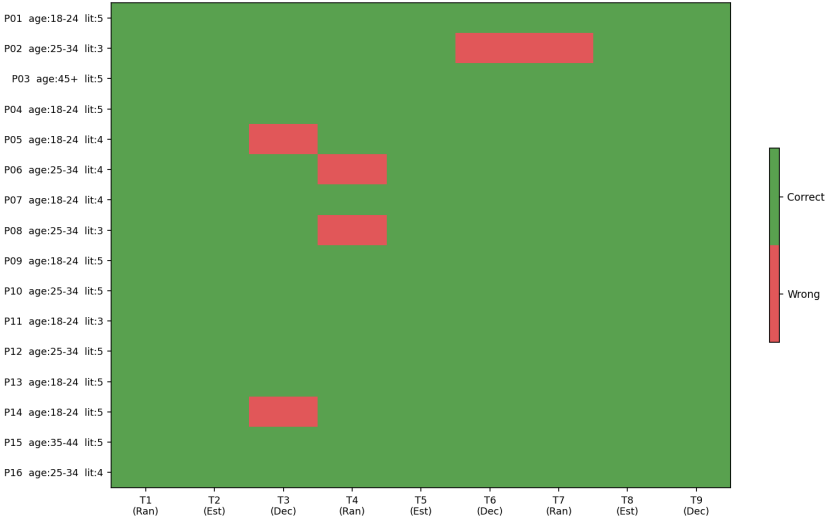


Figure 3: Trial-level correctness heatmap for all 16 participants. Green cells indicate correct answers; red cells indicate errors. Age range and data literacy level are shown on the right axis.

set (for example a dataset with more overlapping distributions, or with a noisy underlying process) would be needed to test H2 on its own terms, because only then would a participant who was genuinely *overconfident* be able to show up in the data.

5.3 Implications

Read cautiously, these results suggest that for simple group-comparison tasks on clean data, a static HOPs grid can match or slightly exceed error bars on accuracy without imposing a clear additional time cost, and that violin/box plots, despite being richer in information, may actively mislead a subset of viewers on ranking tasks where the density silhouette overwhelms the median. None of those statements is strong enough to act on without replication at a larger sample size or with harder task sets, but the direction of the signal is consistent with earlier HOPs work [3] even though our HOPs were static rather than animated.

5.4 Limitations

- *Ceiling effects.* Accuracy was at or near 100% in every encoding, which mechanically compresses differences on both accuracy and Brier score and limits what the omnibus test can see.
- *Small sample.* $n = 16$ is small for Friedman-style tests; effect-size estimates are unstable and the significant omnibus result on accuracy did not survive Bonferroni correction in the post-hoc pairwise tests. We should treat this as pilot-scale evidence.
- *Sample composition.* The convenience sample was recruited from a Bay Area / Silicon Valley peninsula social network with an above-average density of current and former STEM graduate students and industry engineers. Self-rated data literacy was correspondingly high (median = 5/5). This biases performance upward in every condition and is likely part of why the data hit ceiling; a sample drawn from the general public would almost certainly show lower absolute accuracy and, possibly, wider spreads between encodings.
- *Task simplicity.* Three task types on a clean, well-separated dataset with three well-known species. The Adelie mean in particular is so far from the other two that most participants likely did not need any uncertainty information to answer ranking and decision questions correctly.
- *Static HOPs only.* Animated HOPs, which is the original form in [3], were not tested. The static grid is a practical simplification, not a faithful replication of prior HOPs work.
- *Parallel task sets.* Parallel sets reduced answer carry-over but introduced slight content variation between blocks (different ranking targets, different estimation species). Any leftover difficulty asymmetry between sets could show up as encoding difference because of the within-subjects block assignment.
- *Environmental inconsistency.* Most sessions were proctored over Zoom or FaceTime, but a few participants used the hosted link independently. Proctor presence can introduce implicit time pressure, compress reading, and change the set of clarifying questions participants ask. This adds noise to the response-time measure in particular.

5.5 Suggestions for Future Research

- Test animated HOPs against the same baselines to see if motion further improves the advantage seen with the static grid.

- Use more complex tasks (trend detection, outlier identification) where distributional shape should matter more.
- Recruit a larger and more diverse sample, including domain scientists outside of STEM computing fields.
- Test on noisier datasets where group differences are less obvious.
- Explore hybrid encodings that combine error bars with distributional overlays.

6 Conclusion

On common group-comparison tasks over a clean, well-separated dataset, a static HOPs grid produced no observed errors in this sample, while error bars and violin/box plots produced a small handful. In the same sample, error bars were numerically the fastest and best-calibrated encoding, and violin/box carried a descriptive cost on the ranking task and on per-trial response time. Because every encoding sat at or near ceiling, none of these gaps was large enough to survive a conservative statistical threshold once we corrected for multiple comparisons. We therefore avoid framing the 100% HOPs result as “perfect accuracy,” and read it instead as “zero errors observed on these specific tasks, in this specific sample.”

We present these results as pilot-scale, directional evidence rather than as a definitive recommendation. They are consistent with the broader finding in the HOPs literature [3] that showing multiple plausible outcomes can help viewers reason about uncertainty, and they suggest a concrete follow-up: the same design with a harder stimulus set (more overlapping distributions, noisier data), a larger and more diverse sample, and an animated HOPs condition to pair against the static grid.

The study’s primary contribution is methodological: a controlled, within-subjects comparison of three widely used general-purpose uncertainty encodings on three standardized task types, with explicit definitions of *correct*, an overconfidence index alongside the Brier score, and a participant-level, tie-corrected non-parametric analysis. That combination fills a gap in the recent literature, which has focused on formal frameworks, modality comparisons, or domain-specific solutions rather than on head-to-head empirical tests of the general-purpose encodings that most practitioners actually reach for.

Closing Thoughts

Uncertainty communication sits at the intersection of statistics, perception, and design, and most scientific figures still default to a simple error bar. Even

at this pilot scale, the data suggest the default is not automatically the best choice, and that encoding decisions are empirical questions as much as aesthetic ones. Scaled up to harder tasks and a more diverse sample, this line of work could produce practical guidance that meaningfully improves how uncertainty reaches the readers of scientific figures.

Acknowledgments

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